



Princeton Computer Science Contest – Fall 2025

Problem 4: Virtual Memory (18 points)

By Jishnu Roychoudhury

Problem Statement

You're given a virtual memory system with a page table that can hold C page frames, starting empty, and a reference sequence R . Each reference is a hit if the referenced page is already loaded in a frame; otherwise it is a page fault (the page must be brought into memory). If all frames are occupied when a page fault occurs, the system must evict exactly one resident page (no prefetching or bypassing is allowed).

Your tasks:

- Determine the minimum possible number of page faults over the whole sequence.
- Determine the number of eviction decision sequences that achieve this minimum.

Background: Virtual Memory and Paging

A typical modern computer might have 32GB of RAM and 512GB of storage. However, programs cannot use the storage drive as if it were regular memory. Because programs often need to work with more data than can fit into RAM at once, computers use a technique called **virtual memory**.

Virtual memory makes it look as though each program has access to a very large amount of memory, even if the computer does not actually have that much RAM. The computer's operating system does this by moving pieces of a program's memory back and forth between RAM and the storage drive as needed.

The standard technique for managing this process is called **paging**. In paging, both virtual memory and physical memory are divided into small, fixed-size blocks.

- A block of virtual memory is called a **page**.
- A block of physical memory (RAM) is called a **frame**.

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Pages and frames are the same size, which means any virtual page can be loaded into any physical frame. The operating system keeps track of which virtual pages are currently in RAM using a data structure called the **page table**.

Whenever a program needs to access a piece of data, the hardware checks whether the corresponding page is already stored in a frame:

- If the page is in RAM, the access is a **page hit**, and the program continues normally.
- If the page is not in RAM, a **page fault** occurs. The operating system must pause the program, load the needed page from storage into a frame, and then resume execution.

If there is an empty frame available when a page fault happens, the system simply places the new page there. But if all C frames are already full, the operating system must select one of the pages currently in RAM to remove in order to make space for the new one. This process is called **page eviction**. Different **page-replacement policies** define how the system chooses which page to evict.

As a program runs, it produces a sequence of page accesses called a **reference sequence**. For each page in the sequence, the system checks whether the page is currently loaded, and then records either a hit or a fault depending on the contents of the frames.

In the simplified model used in this problem, we assume that physical memory is **fully associative**. This means that any page can be placed into any of the C available frames; there are no restrictions on where pages may go. This setup reflects how paging typically works in real operating systems.

Tests

You are given three different reference sequences and capacities. You get 3 points for solving each task for each reference sequence, for 6 points per reference sequence.

The reference sequences are the following:

- $R = [7, 0, 1, 2, 0, 3, 0, 4, 2, 3, 0, 3, 2]$, $C = 3$. (Hint: 7 misses is the best possible!)
- $R = [5, 3, 2, 2, 1, 1, 2, 0, 0, 3, 0, 2, 2, 4, 2, 0, 5, 3, 4, 0, 5, 2, 1, 2, 5, 4, 4, 4, 2, 0, 3, 0, 1]$, $C = 3$.
- See the file 'seq3.txt' for the access sequence. $C = 4$.

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How to Submit: Submit single integers into our Google forms. We allow a total of 3 submissions for each of the three reference sequences. Note that you can submit to each “task” separately, if you wish (e.g., if you only answer the first task and leave the second one blank, we will count this as half a submission).

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